

Southern Technical University

Engineering Technical College/Basra

Department of Electrical Power Engineering

Final Exam (2024-2025)/A

Class: Second Year

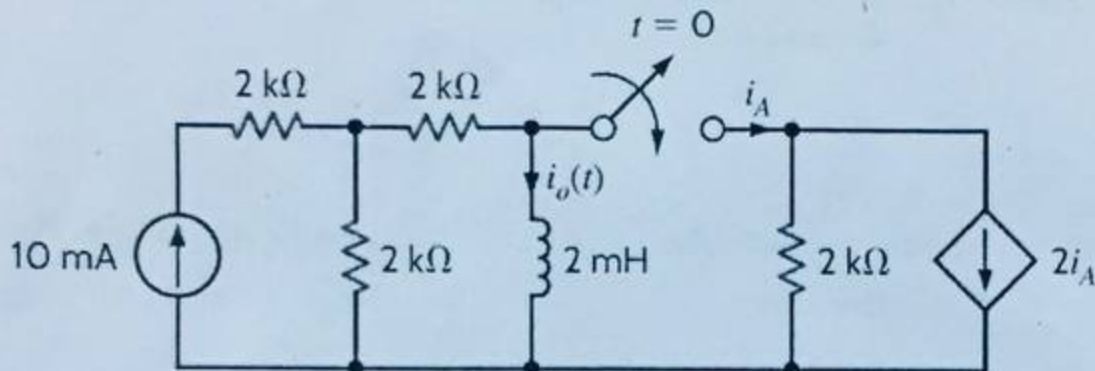
Subject: Electrical Circuit Ana.

Time: 2 Hours

Date: 17 /12 / 2024

(All questions have the same marks)

Q1: Find $i_o(t)$ for $t > 0$ in the following network.

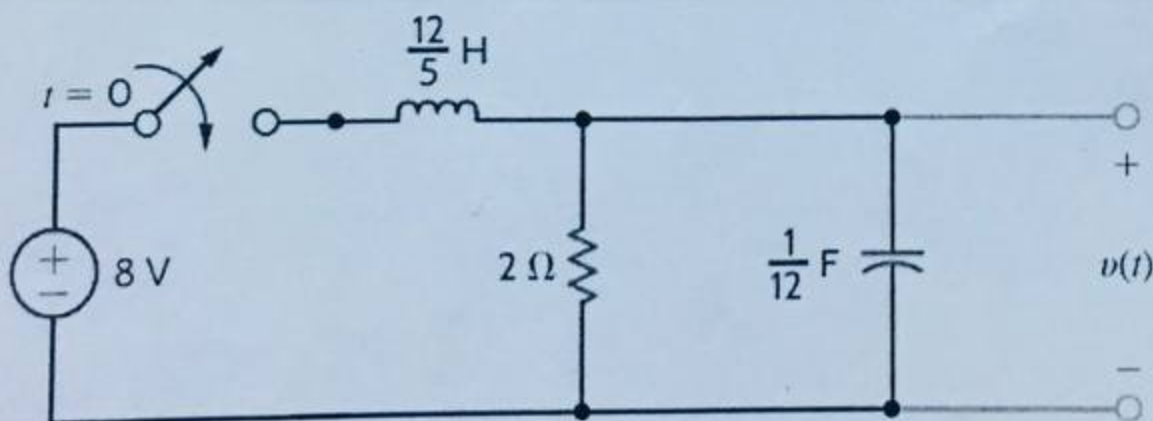


Q2:

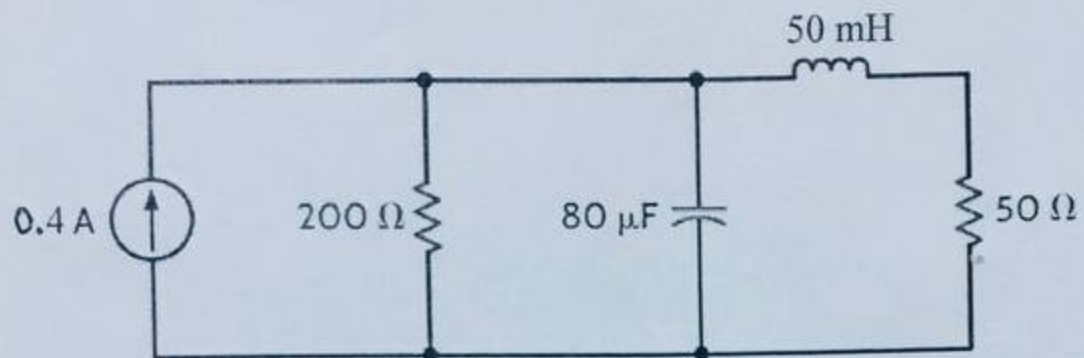
Three branch currents in a network are known to be $i_1(t) = -8 \sin(1000t - 120^\circ) A$, $i_2(t) = -6 \cos(1000t - 60^\circ) A$, and $i_3(t) = -12 \sin(1000t + 45^\circ) A$ by using the three condition details and also by drawing the corresponding signals, determine:

1. The phase angles between $i_1(t)$ and $i_2(t)$.
2. The phase angles between $i_2(t)$ and $i_3(t)$.

Q3: In the circuit shown, find $v(t) > 0$.



the energy stored in the capacitor and inductor.



Good Luck

Doaa Hakim
Examiner

Abdul Hasan Fadhil
Examiner

Dr. Hayder Almukhtar
Head of Department

Southern Technical University

Engineering Technical College/Basra

Department of Electrical Engineering Tech.

Final Exam- 1st Semester (2024-2025)

Class: 2nd Year

Subject: Electronic / 1

Time: 2 Hours

Date: 2/1/2025

((Answer only four questions))

Q1: A crystal diode having internal resistance $r_f = 20 \Omega$ is used for half-wave rectification. If the applied voltage $(t) = 50 \sin \omega t$ V and load resistance $R_L = 800 \Omega$, find :

- (1) $I_m, I_{dc}, I_{r.m.s.}$
- (2) AC power input and DC power output.
- (3) DC output voltage.
- (4) Efficiency of rectification.

(25 Marks)

Q2: A- Find the average and r.m.s values of the given voltage waveform shown in Fig. 1.

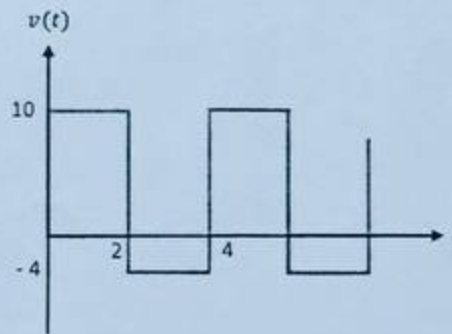


Fig. 1

(15 Marks)

B- Draw the block diagram of the DC power supply.

(10 Marks)

Q3: For the circuit shown in Fig. 2, find: (i) the output voltage (ii) the voltage drop across series resistance (iii) the current through Zener diode.

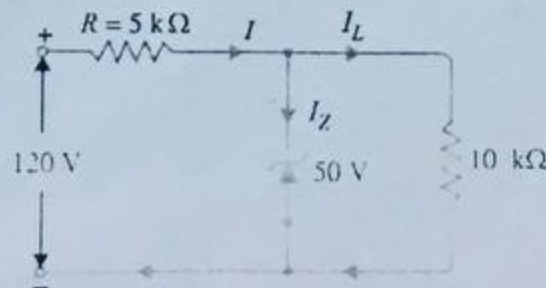


Fig. 2

(20 Marks)

Q4) Given the fixed bias circuit shown in Fig. 3, with $V_{CC} = 20\text{ V}$, $\beta = 150$, $V_{CE} = 4.49\text{ V}$, and $I_C = 10.339\text{ mA}$. Determine the values of R_C and R_B .

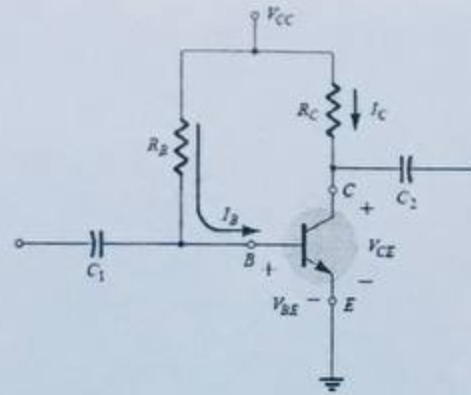


Fig. 3

(25 Marks)

Q5) Given the load line in the Fig. 4 shown below and the defined Q-point, determine the required values of V_{CC} , R_C , and R_B for a fixed-bias configuration.

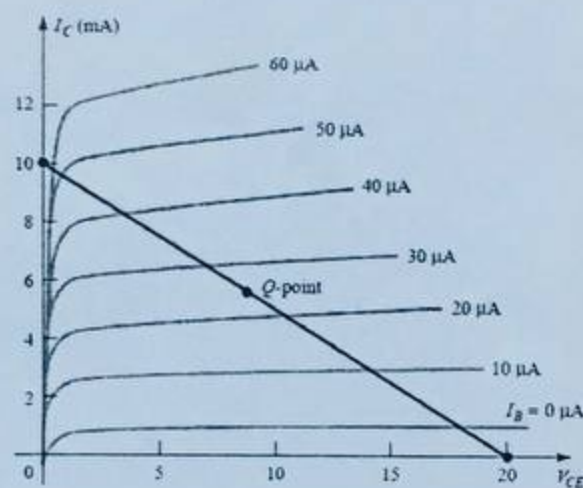


Fig. 4

(25 Marks)

Good Luck

Dr. Mohammed Alkhafaji
Examiner

Dr. Hayder Almukhtar
Head of Department



Southern Technical University

Engineering Technical College/Basra

Department of Technical Electrical Engineering

Final Exam of First Semester (2024-2025)

اسم الطالب كبراهيم محمد طرايع :

Class: Second Year

Subject: computer programming/1

Time: 2 Hours

Date: 24 /12 / 2024

Answer all questions

Q1: Select the correct answer.(20 marks)

1. Who developed Visual Basic? a) Symantec b) Ashton-Tate c) Microsoft d) Sybase.
2. Which of the following extensions is used to represent the project file in Visual Basic?
a) .vbp b) .vb c) .cls d) .vvb
3. Which of the following applications can be developed using Visual Basic tool?
a) Graphical User Interface b) Real-time c) Character User Interface d) All of the mentioned.
4. In the Visual Basic application, which of the following are listed in a properties window?
a) Items b) Values c) Attributes d) Objects.
5. Which of the following is the default name assigned to the label control in Visual Basic?
a) Label1 b) Default Label c) Label0 d) New Label.
6. Which of the following windows is the central to the development of Visual Basic applications
a) Project window b) Form window c) Properties window d) All of the above.
7. Which of the following provides quick access to commonly used commands in the programming environment
a. Toolbox b. Object browser c. Toolbar d. None of the above.
8. is used for finding out about objects, properties and methods. a) Object browser
b) Property window c) Form layout window d) Code editor window.
9. The form module has file extension.... a) .frb b) .fra c) .frm d) .fru.
10. control is used to display text, but user cannot change it directly. a. Textbox
b. Label box c. List box d. Command button.
11. What is the output of the following code `MyString="Hello World"`
`LeftString=left(MyString,3)` a. Hel b. Hello World c. World d. Wor.
12. method is used to delete items from a list box. a. Remove b. Clear c. Remove item
d. Delete.
13. property is an integer value corresponding to the position of the scroll box in the scroll bar. a. Value b. Change c. Visible d. None of the above.
14. A..... displays a list of items from which user can select one or more items. a. Combo box
b. List box c. Check box d. Scroll bar.

15. The default property for a text box control is..... a) Text b) Password char c) Multiline d) Visible.
16. property is used to return sets or a value that determines whether the control can respond to user-generated events. a) Auto size b) Enable c) Visible d) Value.
17. indicates whether a particular condition is on or off. a) Combo box b) List box c) Check box d) None of these.
18. In GUI, is a means of selecting one of several choices. a) Option button b) Scroll bar c) List box d) Combo box.
19. property designates the text appearing in the form's title bar. a) Text b) Name c) Caption d) None of these.
20. The method, which loads the form into memory and displays it on screen a) Load b) Show c) Display d) All of these

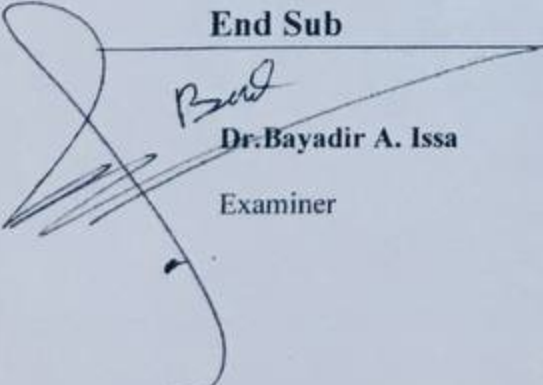
.....
Q2: Complete the following. (10 marks)

- 1- is a list of the forms and modules for the current projects.
- 2- is the most important visible object, without it no control can be displayed.
- 3- Used as a switch (such as OK and Cancel) buttons. Code is written in the Click event procedure of this control.
- 4-: a special List Box filled automatically with names, of the files in a specified directory. It is a list invariant.
- 5- The input box and message box used to input and output variable or message. Both needs and appear at run time.

.....
Q3: Answer with true or false for the following codes and correct the wrong code and remember each code used for what? .(10 marks)

- 1- Private Sub Command1_Click()
Timer1.Enabled = True
End Sub
Private Sub Timer1_Timer()
t = Rnd * 15
Label1.ForeColor = QBColor(CInt(t)).
- 2- MsgBox "are you sure you want to delete this file",vbcritical+vbYesNo,"delete"
- 3- Private Sub Check2_Click()
Txt.FontItalic = Check2.Value
End Sub.
- 4- Txt2.Text = Text1.Text.
- 5- Private Sub Command1_Click()
X=Text1.Text
MsgBox (welcome)
Text1.BackColor = QBColor(9)
Text1.Text = " "
End Sub

Good Luck


Dr. Bayadir A. Issa

Examiner


Dr. Hayder Almukhtar

Head of Department

اسم الطالب: كبرياء كاظم



المادة : جرانم حزب البعث

المرحلة: الثانية

الوقت: ساعتان

التاريخ ٢٠٢٤/١٢/٢٦

وزارة التعليم العالي والبحث العلمي

الكلية التقنية الهندسية/ البصرة

قسم هندسة القوى الكهربائية

عرف

س١/ ١- جرانم السلطة والحكومة ٢- مقبرة شهداء السلام ٣- الجرانم البيئية ٤- مقبرة سيد منيهل
٥- معركة نهر جاسم؟ (١٠ درجات)

س٢/ املأ الفراغات الاتية:

- ١- ابرز ادوات الدولة فاعلية في صيانة حقوق الانسان وتعزيدها هو
 - ٢- اعتمد النظام البعثي منذ تسنمه مقاليد الحكم على تعبئة الجماهير اذ كانت هناك جملة من الأهداف التي تصب في مصلحة النظام منها تنظيم مؤسسات رديفة للجيش تقوم على تنظيمات يقودها الحزب مثل و.....
 - ٣- شهدت محافظة البصرة التي كانت مقصداً للسانحين ورجال الاعمال والتجار من خارج العراق اكبر عملية للإبادة البيئية والبشرية نتيجة امرين و.....
 - ٤- توصف المقابر الجماعية.....
 - ٥- كانت مقبرة جامعة الحلة الدينية في محافظة بابل تمتاز بالخصوصية وذلك لسبب و.....
- (١٠ درجات)

س٣/ تعد سياسة الأرض المحروقة إحدى الطرائق والوسائل البشعة التي تم إتباعها لتدمير بيئة العراق وهي في الأساس مصطلح عسكري يقوم على إحراق كل ما يمكن أن يفيد منه العدو في عملياته العسكرية ومن احد طرق سياسة الارض المحروقة هو قصف العتبات المقدسة والمساجد والحسينيات؟ وضح ذلك بالتفصيل الممل.

(١٠ درجات)

س٤/ عدد اهم نتائج انتفاضة صفر (انتفاضة أربعينية الإمام الحسين (عليه السلام) في العام ١٣٩٧هـ.

(١٠ درجات)

س٥/ وضح العلاقة بين الانتفاضة الشعبانية والمقابر الجماعية ؟

(١٠ درجات)

لكل مجهود نصيب

رئيس القسم
د. حيدر المختار

استاذ المادة
م.م محمد سلمان كاظم

Answer all questions

Q1)

(30 mark)

If the matrix $A = \begin{bmatrix} -1 & 0 \\ 2 & 3 \end{bmatrix}$, calculate the following:

- 1- $2A + A^T$
- 2- cofactor matrix of A
- 3- If $(A^2 = m A + p I)$, find the value of m and p. (I is identity matrix)

Q2)

A)

(20 mark)

For the vectors $M = i + 2j - k$ and $N = -i + j - k$, determine the following:

- 1- The angle between two vector use dot product.
- 2- The projection vector of N on M.

B)

For $Z = 1 - i$, using De Moivre's theorem to find Z^6 .

(10 mark)

Q3)

A)

(25 mark)

Find the Equation of plane whose pass through point Q(0,-1,0) and P(1,1,1)
and perpendicular to plane $2x + y - 6z = 13$

B)

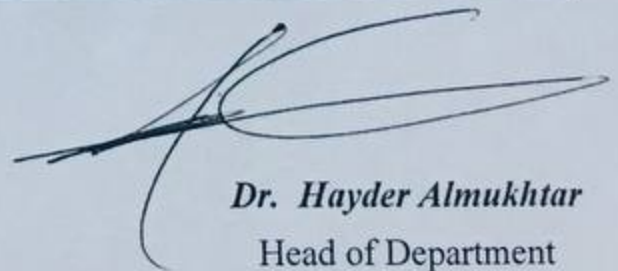
Find the value of x and y that satisfy the equation:

(15 mark)

$$(x-i)^2 - i(2-y) = (1-i)^2$$



Dr. Risala Abuleilah Mohammed
Examiner



Dr. Hayder Almukhtar
Head of Department

Southern Technical University

Engineering Technical College/Basra

Department of Electrical Power Engineering

Final-term Exam (2024-2025) -A

Class: **Second Year**

Subject: **D.C Machine**

Time: **2 Hour**

Date: **31 /12 / 2024**

Answer all questions

Q1: A 4-pole d.c. generator has 600 conductors on its armature. The flux per pole is 0.02 Wb .Calculate the speed at which the generator must be run to generate 300 v, When the generator is connected:

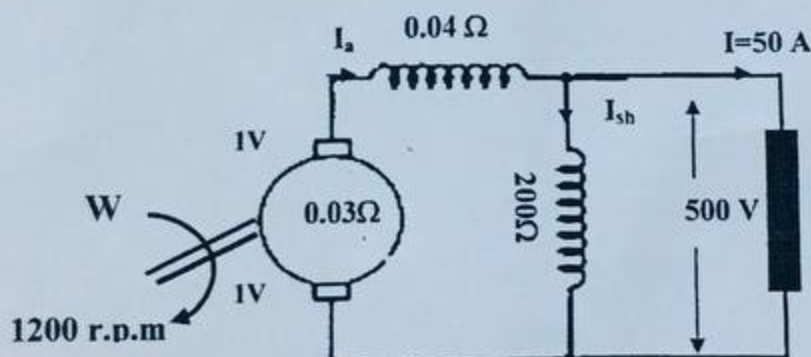
1- Lap - wound . 2- wave-wound. Support your answer with a drawing.

(25 Marks)

Q2: 1- What Types of D.C Motors ? 2- What are the Methods of Speed Control in a DC Motor? Support your answer with a drawing.

(25 Marks)

Q3:



For the circuit as shown above: 4-pole,lap wound, flux per pole (0.02 wb), Determine the e.m.f generated. Calculate also the no.of conductors.

(25 Marks)

Q4: A shunt machine has $R_a(0.04\Omega)$ and $R_{sh}(100\Omega)$ at 460 V Supply and 600 rpm as a generator,50kw.Calculate the speed when running as a motor and 50kw from the same supply.

(25 Marks)

Good Luck

Ms.e Husam Jawad Ali
Examiner

Dr. Hayder Almukhtar
Head of Department